Full measured-boundary FRF comparison (Welch H1 reference)

Dense model evaluation includes all per-input/per-output additive delay phases. Experimental FRF remains an estimate.

Experimental FRF (validation Welch H1) — Before repair — After repair

$u_1 \to y_1$ $u_2 \to y_1$ $u_3 \to y_1$ $u_4 \to y_1$

$u_1 \to y_2$ $u_2 \to y_2$ $u_3 \to y_2$ $u_4 \to y_2$

$u_1 \to y_3$ $u_2 \to y_3$ $u_3 \to y_3$ $u_4 \to y_3$

$u_1 \to y_4$ $u_2 \to y_4$ $u_3 \to y_4$ $u_4 \to y_4$

FRF magnitude (dB, recorded output / motor-command unit)

Frequency (Hz)